SCHOOL OF INFORMATION TECHNOLOGY

RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA

A SYNOPSIS OF MINOR PROJECT

ON

FLIPFOCUS: IOT-BASED LIVE CLOUD ANALYTICS PLATFORM

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BACHELOR OF TECHNOLOGY

Artificial Intelligence and Machine Learning

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Introduction:

The FlipFocus Analytics Hub is an integrated Internet of Things (IoT) hardware and web-based software solution designed to enhance student productivity through tactile interaction. The physical infrastructure utilizes an ESP32 microcontroller paired with a B10K potentiometer dial to allow users to switch dynamically between localized academic profiles.

The system leverages cloud database architectures to manage real-time tracking data without relying on legacy processing components. To ensure immediate feedback and high usability, the device embeds direct audio-visual alerts alongside an on-board character display system. This project provides a robust, cross-platform productivity tracking framework optimized for distraction-free performance in strict institutional environments.

Rationale:

The modern student environment suffers from high digital fragmentation, creating a substantial demand for minimalist, physical hardware interfaces that isolate workflow tracking from distracting smartphone screens. However, generic hardware timers operate on local sandboxes with significant constraints, including a total lack of persistent historic reporting, manual logging requirements, and single-device dependency.

This project explicitly addresses these design constraints by engineering an automated, edge-to-cloud productivity hub. By moving state variables instantly to a secure remote cloud tier, the platform eliminates local memory limitations and drives a separate, user-facing analytics visualization script, making it a highly useful tool for long-term workflow auditing.

Objectives:

- To design and fabricate an IoT edge hardware device using the ESP32 platform to partition focus states dynamically.
- To implement local, high-precision timing logic structures to track accumulated intervals across explicit profiles.
- To provide an aesthetic, glassmorphic web dashboard interface for users to monitor live metrics and active state variables.
- To integrate a real-time server database platform to guarantee instantaneous data synchronization over persistent cloud streams.

Feasibility Study:

The comprehensive implementation feasibility study evaluates technical, operational, economic, legal, and security dimensions:

1. Technical Feasibility:

The core of the project leverages well-established hardware and cloud engineering technologies, including the ESP32 system-on-chip framework, asynchronous HTTPS data stream protocols, and standard modern web rendering engines, which have proven highly effective for decentralized data-logging devices.

Existing cloud database environments (e.g., Google Firebase Realtime Database) and embedded Wi-Fi network stacks provide a solid, low-latency foundation for device development. Compiler toolchains, standard driver packages, I2C abstraction layers, and established serialization libraries are readily accessible worldwide, rendering the engineering cycle fully viable.

2. Operational Feasibility:

- The physical design uses an easy-to-use, analog-to-digital switching dial mechanism where users change states via immediate physical rotation, eliminating complex multi-button setups and ensuring a clean user experience.
- By establishing direct edge-to-cloud synchronization paths, operational points of failure like intermediary gateways or desktop terminal dependencies are bypassed, significantly enhancing system uptime and link availability.
- The web monitoring interface is designed to function cleanly on standard, cross-platform web layout runtimes, allowing it to integrate seamlessly into strict institutional environments, such as smart classrooms or library study networks.

3. Economic Feasibility:

- The edge-computed data-logging model minimizes components by utilizing on-chip processing and open cloud logic tiers, rendering it highly cost-effective compared to traditional commercial tracking hardware.
- By choosing cloud-based serverless datastreaming nodes, ongoing server management and localized hardware maintenance overheads are minimized, allowing the end system to provide maximum analytical utility at near-zero operating costs.

4. Legal and Security Feasibility:

- Local network transmission configurations are enhanced through secure token authentication methods, ensuring the system remains protected against unauthorized data manipulation or session hijacking.
- The remote cloud configuration implements secure data isolation rules, ensuring that interaction records are structured into immutable paths to guarantee transaction privacy and long-term storage integrity.

Methodology / Planning of Work:

The structured engineering layout comprises the following strategic execution steps:

1. **Literature Review:** A comprehensive literature review will be conducted to analyze existing time-tracking hardware limitations and evaluate low-latency IoT transport layers.
2. **System Design:** The system will be designed using a strict modular approach, separating the physical device layout into distinct modules for data acquisition, visual output, audio indicator routing, and cloud sync loops.
3. **Development:** The complete application stack will be developed using a combination of embedded programming languages like C++ for microcontroller firmware and vanilla JavaScript for frontend visualization scripts.
4. **Testing:** The complete hardware-to-cloud lifecycle will be tested thoroughly to verify debounced dial switching, network drop recovery, and latency constraints under varied workloads.

Software/Hardware Required:

- **Programming Languages:** C++ (Arduino Core) and JavaScript (ES6+ Standards).
- **Frameworks & Libraries:** Firebase ESP Client (Mobizt Core), LiquidCrystal_I2C Library, and Native Browser DOM API.
- **Cloud Platform:** Firebase Realtime Database Instance (asia-southeast1 Region node).
- **Hardware Configuration:** ESP32 DevKit V1 Board, B10K Potentiometer, I2C 16x2 LCD Display, HYDZ Buzzer, 5V LED Indicator, and an Intel Core i5/8 GB RAM/256 GB SSD engineering PC.

Expected Outcomes:

The fully realized implementation of the **FlipFocus** ecosystem guarantees the following technical deliverables:

- A highly stable, responsive IoT productivity cube utilizing hardware debouncing and local interval computation logic.
- An aesthetic, low-latency, responsive frontend web interface tracking operational durations without manual polling requirements.
- Isolated profile data slots tracking specific categorical sessions ("Assignment Work", "Exam Prep", "Core Coding", and "Total Break") with smart history defaults.
- Integration with a scalable real-time cloud data structure to ensure persistent, tamper-proof system diagnostics over secure sockets.

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